

Appendix C: Volatility 3 Command Cheat Sheet

This cheat sheet summarises key commands and scripts used in the Memory Forensics and Volatile Artefact Analysis project. It serves as a quick reference for the setup of the environment to handle a memory image and execute the plugins provided by Volatility 3.

Setup the Environment

Purpose	Command
Update system	<code>sudo apt update && sudo apt upgrade -y</code>
Install dependencies	<code>sudo apt install python3 python3-pip git -y</code>
Clone Volatility 3	<code>git clone https://github.com/volatilityfoundation/volatility3.git</code>
Install requirements	<code>pip3 install -r requirements.txt</code>
Verify installation	<code>python3 vol.py -h</code>

Memory Acquisition & Integrity

Purpose	Command
Capture RAM (FTK Imager GUI)	File → Capture Memory
Generate SHA-256 hash	<code>sha256sum [memfile] > ./Desktop/[filename].sha256</code>
Hash-Verify	<code>sha256sum -c ./Desktop/[filename].sha256</code>

System Information

Purpose	Command
OS details	<code>python3 vol.py -f [image] windows.info</code>
Registry hives	<code>python3 vol.py -f [image] windows.hivelist</code>
Version info	<code>python3 vol.py -f [image] windows.verinfo</code>

Process & Execution

Purpose	Command
List active processes	<code>python3 vol.py -f [image] windows.pslist</code>
Scan hidden processes	<code>python3 vol.py -f [image] windows.psscan</code>
Process tree	<code>python3 vol.py -f [image] windows.pstree</code>
Executed commands	<code>python3 vol.py -f [image] windows.cmdline</code>
Loaded DLLs	<code>python3 vol.py -f [image] windows.dlllist</code>

Artefacts of networking

Purpose	Commands
Active connections	<code>python3 vol.py -f [image] windows.netscan</code>
Network sockets	<code>python3 vol.py -f [image] windows.sockets</code>
Environment variables	<code>python3 vol.py -f [image] windows.envvars</code>

Credentials & Security

Purpose	Command
Dump password hashes	python3 vol.py -f [image] windows.hashdump
LSA secrets	python3 vol.py -f [image] windows.lsadump
Cached credentials	python3 vol.py -f [image] windows.cachedump
User privileges	python3 vol.py -f [image] windows.privileges

File System & Persistence

Purpose	Command
Open file handles	python3 vol.py -f [image] windows.filescan
MFT scan	python3 vol.py -f [image] windows.mftscan
Loaded modules/drivers	python3 vol.py -f [image] windows.modules
Callback routines	python3 vol.py -f [image] windows.callbacks
SSDT hooks	python3 vol.py -f [image] windows.ssdtd

Automation Scripts

Script	Example	Purpose
vol3_info.sh	bash vol3_info.sh clean_image.raw	Collects baseline system artefacts
run_vol3_scan.sh	bash run_vol3_scan.sh infected_image.raw	Runs core analysis plugins
vol3_artifacts.sh	bash vol3_artifacts.sh [image]	Extracts credentials & malware artefacts
vol3_compare.sh	bash vol3_compare.sh clean_image.raw infected_image.raw	Compares clean vs infected images

Helper Commands

Task	Command
Create timestamped output folder	folder outdir="outputs/\$(date -u +%Y%m%dT%H%M%S)"; mkdir -p "\$outdir"
View plugin list	python3 vol.py -l
Search for keyword in results	grep -i "plugx" \$outdir -R
Archive results	tar -czvf "\$outdir".tar.gz "\$outdir"

Reference Note

This cheat sheet for Volatility 3 Command supports reproducibility and traceability, which is important in ensuring that the forensic workflow and the process of artefact recovery can be efficiently replicated in future investigations or audits.